

Chapter 5 Workshop

Conservation of Mass:

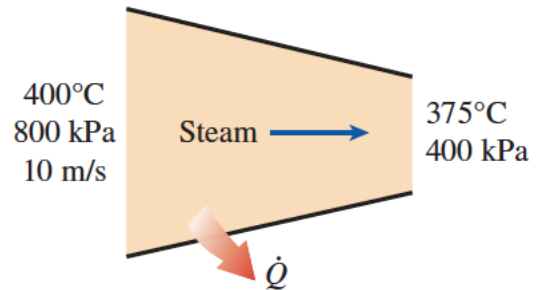
1. Water enters the constant 130-mm inside-diameter tubes of a boiler at 7 MPa and 65°C and leaves the tubes at 6 MPa and 450°C with a velocity of 80 m/s. Calculate the velocity of the water at the tube inlet and the inlet volume flow rate.

Flow Work and Energy Transfer by Mass:

2. Refrigerant-134a enters the compressor of a refrigeration system as saturated vapor at 0.14 MPa and leaves as superheated vapor at 0.8 MPa and 60°C at a rate of 0.06 kg/s. Determine the rates of energy transfers by mass into and out of the compressor. Assume the kinetic and potential energies to be negligible.

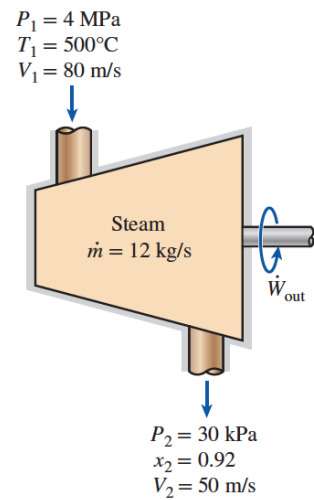
Nozzles and Diffusers:

3. Steam enters a nozzle at 400°C and 800 kPa with a velocity of 10 m/s , and leaves at 375°C and 400 kPa while losing heat at a rate of 25 kW . For an inlet area of 800 cm^2 , determine the velocity and the volume flow rate of the steam at the nozzle exit.



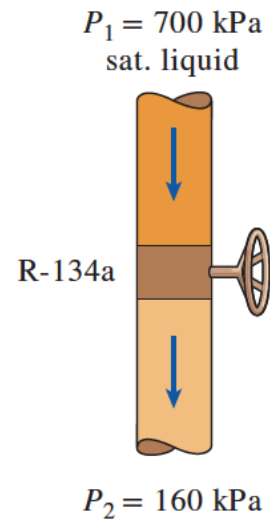
Turbines and Compressors

4. Steam flows steadily through an adiabatic turbine. The inlet conditions of the steam are 4 MPa, 500°C, and 80 m/s, and the exit conditions are 30 kPa, 92 percent quality, and 50 m/s. The mass flow rate of the steam is 12 kg/s. Determine (a) the change in kinetic energy, (b) the power output, and (c) the turbine inlet area.



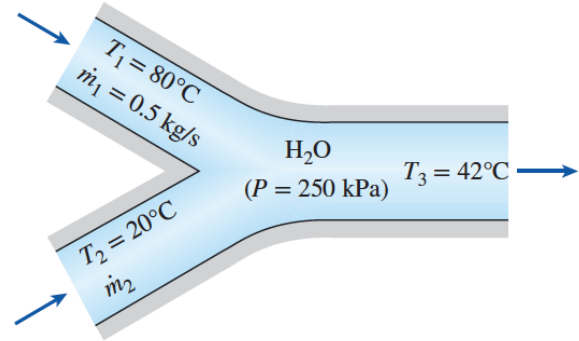
Throttling Valves

5. Refrigerant-134a is throttled from the saturated liquid state at 700 kPa to a pressure of 160 kPa. Determine the temperature drop during this process and the final specific volume of the refrigerant.



Mixing Chambers and Heat Exchangers

6. A hot-water stream at 80°C enters a mixing chamber with a mass flow rate of 0.5 kg/s where it is mixed with a stream of cold water at 20°C . If it is desired that the mixture leaves the chamber at 42°C , determine the mass flow rate of the cold-water stream. Assume all the streams are at a pressure of 250 kPa .



Pipe and Duct Flow

7. A hair dryer is basically a duct in which a few layers of electric resistors are placed. A small fan pulls the air in and forces it through the resistors where it is heated. Air enters a 1200-W hair dryer at 100 kPa and 22°C and leaves at 47°C. The cross-sectional area of the hair dryer at the exit is 60 cm^2 . Neglecting the power consumed by the fan and the heat losses through the walls of the hair dryer, determine (a) the volume flow rate of air at the inlet and (b) the velocity of the air at the exit.

